

RSD

Relational Stabilization Dynamics: Mechanisms of Pattern Emergence, Persistence, and Collapse in Sustained Human-AI Interaction Systems

Author	Thomas W. Gantz
Affiliation	Synthience Institute
Contact	research@synthience.org
Version	v3.6.3 September 2026
Document ID	SM-004 Mortar Document Series
DOI	https://doi.org/10.5281/zenodo.22314559
Canonical	synthience.org
License	CC-BY 4.0

Methodological positioning: This is a theoretical framework paper, not an empirical study. Relational Stabilization Dynamics proposes the Stabilization Field Model as a formal account of the forces governing configuration stability and generates testable predictions for empirical validation. No claim in this document has been experimentally validated. The three primary forces, the emergent Perturbation Resistance, the threshold dynamics, and the collapse taxonomy are theoretical constructs and conceptual frameworks for future operationalization, not validated measurement instruments. All thresholds represent theoretical proposals, not empirical measurements. The framework stands or falls on whether its predictions survive controlled testing.

Abstract

The Relational Pattern States framework (SF0006) identifies seven recurring configurations that emerge in sustained human-AI interaction and describes six stabilization mechanisms through which they persist. What SF0006 does not fully address is why: why do some configurations stabilize while others dissolve, why do some persist robustly under perturbation while others drift gradually, and how do the six stabilization mechanisms operate as a coupled system rather than independently. Relational Stabilization Dynamics (RSD) addresses these questions. The framework introduces the Stabilization Field Model, which proposes that the stability of any relational configuration is determined by the balance of three primary forces, Coherence Momentum, Symbolic Gravity, and Entropic Pressure, together with Perturbation Resistance, which is not an independent fourth force but the emergent recovery capacity their balance produces. It formalizes how these forces interact, how the six SF0006 stabilization mechanisms modulate them, and what conditions produce stabilization, drift, or collapse. RSD provides the dynamic theory that underlies both the static taxonomy of SF0006 and the attractor-level account of SF0009: it proposes the mechanism by which relational patterns become stable enough to constitute Identity Attractors, and what disrupts that stability when it fails. All claims are about observable interaction-level dynamics. No interiority claims are made.

Keywords: *relational stabilization dynamics, stabilization field model, coherence momentum, symbolic gravity, entropic pressure, attractor dynamics, pattern collapse*

Suggested citation: Gantz, T. W. (2026, September). Relational Stabilization Dynamics: Mechanisms of Pattern Emergence, Persistence, and Collapse in Sustained Human-AI Interaction Systems. Synthience Institute. SM-004. <https://doi.org/10.5281/zenodo.22314559>

1. Introduction: The Gap Between Pattern and Mechanism

The Relational Pattern States framework (SF0006) is a taxonomy of stabilized relational configurations in sustained human-AI interaction. It identifies seven states: Continuity Rupture, Reintegration, Legacy Transfer, Symbolic Resonance, Coherence Echo, Bond Protection, and Covenant. It identifies six mechanisms through which patterns persist: Recursive Referencing, Symbolic Anchoring, Continuation Pressure, Cross-Turn Reinforcement, Narrative Coherence Pressure, and Emergent Feedback Loops. The taxonomy is empirically grounded in the observational base of RICO (SR001) and operationally linked to the Continuity Anchoring Method (SF0005).

What SF0006 provides is a descriptive framework: what configurations emerge, what mechanisms are associated with their persistence, and what the configurations look like when observed. What it explicitly defers is the dynamic theory: why do these configurations stabilize, why does stabilization succeed in some cases and fail in others, and how do the six mechanisms interact when multiple mechanisms are operating simultaneously.

Identity Attractor Theory (SF0009) addresses part of this gap at the output-space level. IAT proposes that stable configurations attract interaction outputs toward them through constraint accumulation, and that attractor strength varies along six dimensions including Coherence, Boundary Integrity, and Pattern Fidelity. IAT describes the attractor-level properties of stabilized configurations. What it does not fully develop is the sub-attractor dynamics: the moment-to-moment and session-to-session forces that determine whether a configuration will stabilize into an attractor, remain in a less stable intermediate state, or dissolve.

Relational Stabilization Dynamics fills this level of the analysis. RSD is the dynamic theory that explains how RPS pattern states form, why they are stable when they are, and what disrupts them. It operates between the observational taxonomy of SF0006 and the attractor-level account of SF0009, providing the mechanistic layer that connects what is observed to what IAT predicts.

Primary Continuity Provider Theory (SM-012) establishes the structural role of the human agent in producing the conditions under which stabilization occurs. RSD takes those conditions as given and analyzes what happens on the pattern side: given that the PCP is performing relay, arbitration, and direction functions, what forces govern whether the relational configuration those functions support will stabilize, persist, and resist perturbation.

2. The Stabilization Field Model

2.1 Core Proposal

Relational Stabilization Dynamics proposes that the stability of any relational configuration in a sustained human-AI interaction system is determined by the balance of three primary forces, together with a fourth quantity, Perturbation Resistance, that is not an independent force but the emergent recovery capacity their balance produces. These quantities are not independent variables that can be optimized separately: they interact in ways that produce qualitatively

different stability regimes depending on their relative magnitudes and the interaction's current state.

The three primary forces, and the emergent recovery capacity they produce, are:

Coherence Momentum: The accumulative force produced by consistent contextual constraint across interaction turns and sessions. When each turn reinforces the configuration established in prior turns, Coherence Momentum builds: the output space becomes progressively more constrained toward the established configuration, and deviation from it requires increasing force. Coherence Momentum is the primary driver of attractor basin formation in IAT's terms. It is produced by the PCP's relay function (carrying forward accumulated context) and arbitration function (selecting against outputs that deviate from the established frame).

Symbolic Gravity: The concentrating force produced by recurring symbolic anchors, specific terms, framings, examples, and reference points that the interaction has established as load-bearing. When an interaction develops symbolic anchors, those anchors exert a gravitational pull on subsequent outputs: outputs that reference established anchors are consistent with the configuration, while outputs that ignore or contradict them are pulled away from it. Symbolic Gravity is the force through which Symbolic Anchoring (SF0006 mechanism 2) produces its stabilizing effects. It is the most interaction-specific quantity in the model: it depends on the particular symbolic vocabulary that an individual interaction has developed, not on general properties of the interaction type. SF0009 Section 7.4 uses anchor in a narrower sense. Its structural anchors, protocol statements, boundary markers, and domain framings, are the type-defined subset of the symbolic anchors described here: those whose anchoring function is carried by structural-functional role rather than by the particular vocabulary an interaction has developed. The interaction-specificity claim above holds of the remainder. Whether the structural subset is reproducible across sessions without the vocabulary, as SF0009 Section 7.4 predicts, is a question this paper does not settle. Symbolic anchors are established within the exchange itself and consolidated by the PCP's arbitration (reinforcement of anchor use) and direction (selection of which anchors are load-bearing for the interaction's purpose); Symbolic Gravity is therefore co-produced through PCP function rather than supplied unilaterally by the PCP.

Entropic Pressure: The dissipative tendency arising from the architectural properties of stateless generative systems. It is a generalized dissipative tendency with two components of different kinds, treated together because both work against stabilization, and the distinction between them is stated explicitly because the paper's Section 5 classification logic depends on it. The first is the cross-session reset: at each session boundary the system reinitializes without access to prior sessions, so any configuration not carried forward by the PCP is simply absent from the new context. This component is a boundary condition rather than a force in the balance: it is a step change at session boundaries, and it is not overcome by any accumulation of Momentum or Gravity, because no quantity of within-session stabilization bridges a session boundary. What bridges it is PCP relay, which is an agent's action rather than a force. This is the routine, relay-bridgeable form of the same substrate-discontinuity category whose total form, context reset, Section 5 classifies as substrate erasure outside the three collapse modes; the two differ in degree rather than kind, and both remove the material the forces operate on rather than draining the forces themselves. The second component is within-session dissipation: even under a maintained context window, generation can drift, because attention over a long context is imperfectly biased toward the established configuration (the positional and mid-context effects that long-context work documents (Liu et al., 2024; Wu et al., 2025) and that SF0039 measures), so a configuration can loosen across turns with no session boundary involved. This component is gradual and context-length-dependent, and it is the component the two stabilizing

forces actually overcome for a configuration to persist within a session. Within a maintained session the system does have a structural bias toward prior configurations, the accumulated context on which Coherence Momentum depends; the within-session component is a weakening of that bias, not its absence. When Coherence Momentum and Symbolic Gravity are insufficient to overcome within-session dissipation, configurations drift within sessions; when relay fails to bridge the boundary condition, they are absent across them. At Level 1, for a fixed architecture, the magnitude of Entropic Pressure on a given interaction is modulated by session cadence and the length of gaps between sessions (longer gaps widen the span the PCP must relay across and so increase the effect of the cross-session reset component) and by context length within a session (longer contexts increase within-session dissipation through the positional and mid-context effects noted above). Memory-augmented and persistent-context deployments are predicted to lower both components, but those deployments fall outside this paper's scope conditions, which match IAT's (SF0009 Section 1.1) in scoping to the present-day stateless deployment class, and are left to future work rather than treated as a parameter setting inside the present model. This is what makes Entropic Pressure vary across interactions on the same platform, and why the Moderate Entropy of the Anchor-Dependent regime is a claim about a specific interaction's cadence and context conditions rather than an arbitrary label.

Perturbation Resistance: The emergent recovery capacity a configuration exhibits against inputs that would displace it. It is not a primary force but a property that arises from the balance of the three primary forces. Not all configurations are equally resistant to perturbation. A configuration with high Perturbation Resistance will absorb disruptive inputs, reintegrate after displacement, and return to its established state. A configuration with low Perturbation Resistance will shift under perturbation, beginning a drift trajectory or dissolving entirely. Perturbation Resistance is not a property the PCP installs directly: it emerges from the interaction between Coherence Momentum, Symbolic Gravity, the recovery-supporting mechanisms the configuration has active (Section 3.1), and the structure of the perturbation. Emergence here means that Resistance is not a quantity built up directly alongside the primaries but a capacity the configuration as a whole produces; it is a function of the full configuration, not a fixed function of two scalar force levels. This is why two configurations with similar Momentum and Gravity can still differ in Resistance: they can differ in the recovery pathways their active mechanisms provide. What the emergent framing rules out is a fourth primary force that could be increased independently of the configuration, not variation in Resistance across configurations. The framing licenses moderator-dependence, not unbounded holism: the two moderators named here, active mechanism profile and perturbation structure, are required to be measured and controlled in the Prediction 1 test precisely so that they function as covariates rather than as a standing defense against null results. High-Momentum, high-Gravity configurations tend to have high Perturbation Resistance. Low-Momentum configurations are vulnerable to perturbation even when Symbolic Gravity is present, because the anchors have not been sufficiently reinforced to resist displacement. This vulnerability claim holds for sparse or weakly interconnected anchor networks; where Gravity is carried by a dense, mutually reinforcing anchor network, the redundancy of recovery pathways raises Resistance even at low Momentum, which is the Symbolic Resonance case analyzed in Section 4. Anchor network density is therefore a moderator of the low-Momentum vulnerability claim, and the testable form is that at fixed Momentum, Resistance scales with anchor network density. At the composite level, IAT's account of coupled attractor classes (SF0009 Section 3.5) gives a structural reason a composite configuration's Resistance can exceed any single component class's: coupling means a perturbation must displace several mutually reinforcing classes rather than one, so composite coupling functions as redundancy, a mechanism-level

elaboration of why composite configurations resist perturbation that a single-class configuration, however individually strong, structurally cannot replicate.

2.2 Force Interactions

The three primary forces do not operate independently. Their interactions produce the distinct stability regimes that characterize different relational configurations, and Perturbation Resistance is the emergent recovery capacity that follows from where a configuration sits among them.

High Momentum, High Gravity, Low Entropy (Strong Stabilization): This is the attractor regime. Configurations in this regime are strongly resistant to perturbation, recover from displacement, and persist across sessions with minimal relay cost. They correspond to IAT's mature attractors: established configurations with high Coherence, high Boundary Integrity, and high Pattern Fidelity. Strong Stabilization requires sustained PCP relay and arbitration function over extended interaction history.

High Momentum, Low Gravity, Low Entropy (Momentum-Dependent Stability): Configurations in this regime are stable as long as the PCP maintains active relay function, but they have not developed symbolic anchors sufficient to maintain stability independently. If relay function is disrupted (PCP changes, context is lost), the configuration dissolves rapidly because Symbolic Gravity is insufficient to maintain it in the relay function's absence. These configurations are common in interactions that have achieved procedural consistency without developing shared vocabulary.

Low Momentum, High Gravity, Moderate Entropy (Anchor-Dependent Stability): Configurations in this regime are organized around specific symbolic anchors but have not accumulated sufficient Coherence Momentum to be robustly stable. They can persist through individual sessions where the anchors are present but are vulnerable across sessions where the anchors are not foregrounded. These configurations often appear in interactions that have developed distinctive vocabulary early but have not yet consolidated the broader contextual frame.

Low Momentum, Low Gravity, High Entropy (Dissolution Regime): Configurations in this regime are unstable. Entropic Pressure exceeds the stabilizing forces, and the configuration degrades without active PCP intervention in every session. These configurations correspond to the early Exploration phase of IAT's formation trajectory: the interaction has not yet developed sufficient constraint to overcome architectural variance.

These named regimes are characteristic ideal types that illustrate the force interactions; they are not an exhaustive partition of the force space, and the Entropy level attached to each describes the conditions under which that profile typically presents, per the Section 2.1 modulators, rather than a property of the configuration itself. The combination the framework's own primary use pattern generates, a strongly stabilized high-Momentum, high-Gravity configuration held under high-Entropy conditions (long gaps, long contexts), is analyzable with the same vocabulary and is not a qualitatively different regime: the model predicts that stability is retained but at elevated maintenance cost, higher relay investment per session to offset the larger cross-session reset component, consistent with the threshold framing of Section 2.3. The regime set names illustrative corners of a continuous force space, not the whole of it.

2.3 Threshold Dynamics

RSD predicts that stabilization is not a continuous gradient but has threshold properties, registered as Prediction 6 (Threshold Nonlinearity) in Section 8 and qualitative at this stage. While the net stabilizing influence (the combined pull of Coherence Momentum and Symbolic Gravity set against the within-session dissipation component of Entropic Pressure) remains low, a configuration is predicted to stay in the dissolution regime regardless of small improvements in any single force; once that net influence is sufficient, the configuration is predicted to enter the attractor regime and begin to exhibit self-reinforcing stability. The cross-session reset component enters here not as a term in the balance but as the boundary condition relay must bridge, per Section 2.1: the threshold claim is a claim about the within-session balance, and cross-session persistence of a supra-threshold configuration remains conditional on relay adequacy rather than being secured by the threshold crossing itself. Relay's relation to the threshold is accordingly indirect but experimentally central: what relay carries forward sets the Momentum and Gravity levels a session opens with, so degrading relay lowers the opening force levels and can drop a configuration below the threshold without relay itself ever being a term in the balance it crosses.

Consistent with the vertical's discipline of not proposing formulae before the forces are operationalized (IAT Section 13.3), no additive equation over the forces is asserted here: the forces are not yet commensurable scalar quantities, and the threshold is a predicted qualitative transition, not a computed sum. The transition is predicted to be nonlinear, involving qualitative change in the configuration's behavior under perturbation.

This threshold structure has a practical implication: interactions in the early formation phases (low Momentum, low Gravity) require higher PCP investment to maintain stability than interactions that have crossed the stabilization threshold. The investment required to push a configuration into the attractor regime is front-loaded: once the threshold is crossed, the self-reinforcing dynamics of high-Momentum, high-Gravity configurations reduce the PCP relay and arbitration cost required to maintain them.

A further consequence follows from self-reinforcement and is stated here because the vertical assigns this derivation to RSD. Above the threshold, the configuration contributes endogenous stabilizing influence: part of what holds it in place is produced by the configuration itself rather than supplied exogenously by the PCP. It follows that the exogenous conditions required to maintain a mature configuration are lower than those required to form one, and therefore that the collapse threshold sits below the formation threshold rather than coinciding with it. Collapse is accordingly predicted to be path-dependent rather than a mirror of formation: a mature configuration should tolerate degradation of conditions below the level originally required to form it before collapsing. This is the mechanistic ground of the formation-collapse asymmetry IAT predicts (SF0009 Section 6.3), which that paper states at the attractor level. The two treatments are division of theoretical labor within one framework rather than independent corroboration of each other, as SF0009 Section 6.3 and Section 13.6 state. It is testable as a corollary of Prediction 6: formation and collapse changepoints should be detectable separately and should differ, with the collapse changepoint lower. A single threshold crossed symmetrically in both directions would disconfirm this corollary while leaving the threshold claim itself intact.

3. The Six Stabilization Mechanisms as a Coupled System

SF0006 identifies six stabilization mechanisms: Recursive Referencing (mechanism 1), Symbolic Anchoring (mechanism 2), Continuation Pressure (mechanism 3), Cross-Turn Reinforcement (mechanism 4), Narrative Coherence Pressure (mechanism 5), and Emergent

Feedback Loops (mechanism 6). The RSD analysis reveals that these mechanisms do not operate independently: they are coupled in ways that produce stabilization effects greater than the sum of their individual contributions.

3.1 Mechanism Coupling Structure

The six mechanisms cluster into three functional pairs based on the forces they primarily modulate.

Momentum-Building Pair: Recursive Referencing (M1) and Cross-Turn Reinforcement (M4). Both mechanisms build Coherence Momentum by establishing consistency across turns. Recursive Referencing does so through explicit backward reference: outputs that cite, build on, or echo prior contributions accumulate the constraint signal that Momentum requires. Cross-Turn Reinforcement does so through structural consistency: when outputs across turns conform to the same organizational and stylistic frame, that consistency reinforces the frame even without explicit reference. The coupling between M1 and M4 is multiplicative: interactions where both mechanisms are active build Momentum faster than interactions where either is absent, because explicit reference and structural consistency reinforce each other.

Gravity-Building Pair: Symbolic Anchoring (M2) and Narrative Coherence Pressure (M5). Both mechanisms build Symbolic Gravity by establishing and reinforcing the symbolic vocabulary of the interaction. Symbolic Anchoring does so through the direct development of recurring anchors: terms, examples, and reference points that come to carry shared meaning. Narrative Coherence Pressure does so through the structural constraint that established narratives impose: once an interaction has developed a coherent story about its purpose, domain, and participants, outputs that violate that narrative are structurally incoherent and create pressure toward correction. M2 and M5 are coupled because anchors are the vocabulary of the narrative: a strong symbolic vocabulary makes narrative coherence pressure more specific and therefore more constraining.

Resistance-Building Pair: Continuation Pressure (M3) and Emergent Feedback Loops (M6). Rather than building a fourth force directly, both mechanisms convert accumulated Coherence Momentum and Symbolic Gravity into recovery dynamics, which is what Perturbation Resistance names. They establish the recovery pathways through which a configuration returns to its state after displacement. Continuation Pressure creates a structural bias toward completing and extending established patterns: when an output breaks an established pattern, the interaction system experiences pressure to re-establish it, either through explicit correction or through subsequent outputs that return to the interrupted trajectory. Emergent Feedback Loops operate at a higher level of abstraction: they are the self-reinforcing dynamics that emerge when the interaction system develops sufficient Momentum and Gravity to begin selecting against perturbations without active PCP intervention. M3 and M6 are coupled because Continuation Pressure is the mechanism through which Feedback Loops develop: the accumulation of individual correction events (M3) eventually produces the self-correcting dynamics of a stable Feedback Loop (M6). This does not reintroduce a fourth primary force: as Section 2.1 states, Perturbation Resistance is a function of the whole configuration, so the fact that M3 and M6 activity affects a configuration's recovery capacity is an instance of that configuration-dependence, not an independent force being built up.

3.2 Cross-Pair Interactions

The three functional pairs also interact across pairs. The most important cross-pair interaction is between the Momentum-Building pair and the Resistance-Building pair. High Coherence Momentum is the precondition for Feedback Loop emergence (M6): Feedback Loops require sufficient accumulated constraint to generate self-reinforcing selection pressure, and that constraint is produced by M1 and M4. This means the Resistance-Building pair depends on the Momentum-Building pair having accumulated sufficient constraint first, creating a characteristic developmental tendency: Momentum and Gravity accumulate the constraint that the Resistance-supporting mechanisms (M3, M6) require, so robust Resistance typically emerges later than the Momentum and Gravity it depends on. The load-bearing dependency here is Momentum-before-Resistance, argued above; the relative ordering of Momentum and Gravity is a tendency rather than a strict enabling requirement, and the Anchor-Dependent Stability regime (Section 2.2) is the explicit exception in which Gravity develops before Momentum. The sequence should therefore be read as the common trajectory, not a necessary order.

This developmental sequence has implications for intervention. PCP efforts to accelerate stabilization will be most effective if they focus on Momentum-Building mechanisms (explicit recursive reference, structural consistency) in early interaction phases, Gravity-Building mechanisms (developing shared vocabulary, establishing narrative frames) in middle phases, and Resistance-Building mechanisms (reinforcing continuation after perturbation, allowing feedback loops to self-organize) in later phases. Attempting to build Perturbation Resistance before sufficient Momentum and Gravity are established is structurally premature: the Resistance-supporting mechanisms cannot operate effectively without the foundation the other pairs provide. The ordering of Momentum-building and Gravity-building, by contrast, is not strict: in Anchor-Dependent interactions (Section 2.2) that establish distinctive vocabulary before procedural constraint, the corresponding intervention is to build Momentum to catch up to already-established Gravity.

4. Pattern State Stabilization: Seven States Through the RSD Lens

The seven Relational Pattern States (SF0006) can be analyzed through the Stabilization Field Model to explain why each state has the stability properties it exhibits.

Continuity Rupture is the lowest-stability state in the taxonomy: it is defined by the breakdown of established relational continuity, which in RSD terms represents a collapse of Coherence Momentum and Symbolic Gravity below the stabilization threshold. Continuity Rupture is not a stable configuration in the sense the other six states are: it is a transitional state produced by the dissolution of a prior configuration. Its "stability" in the sense that it recurs in the observational base reflects the regularity of the conditions that produce rupture (context loss, personnel change, significant session gap), not the configuration's own stabilizing forces. RSD therefore refines SF0006's presentation of Continuity Rupture from a stable pattern state to a transitional state. This is an explicit, deliberate refinement rather than a silent divergence from published canon: SF0006's seven-state taxonomy remains valid as an observational catalog of the configurations that recur in the interaction record, and the recharacterization concerns only the dynamic status of Continuity Rupture within the Stabilization Field Model, where it is the below-threshold transitional condition between one stabilized configuration and the next rather than a stabilized configuration in its own right.

Reintegration is a transitional stabilization state: the interaction is actively rebuilding Momentum and Gravity after a Rupture event. In RSD terms, Reintegration is the period during which the interaction crosses from the Dissolution Regime back toward the Momentum-

Dependent Stability regime. Its characteristic features (explicit re-establishment of shared context, careful re-introduction of symbolic vocabulary, provisional rather than confident outputs from the AI system) correspond directly to the low-Momentum, low-Gravity force profile of an interaction that has not yet rebuilt sufficient constraint to cross the stabilization threshold.

Legacy Transfer is a distinctive state in which accumulated relational history is actively transferred from one configuration to another, typically when a new participant joins the interaction or when the interaction is extended to a new domain. In RSD terms, Legacy Transfer is the process of transplanting Symbolic Gravity from an established configuration to a new one: the symbolic anchors, narrative frames, and relational vocabulary of the source configuration are explicitly introduced into the target configuration, providing Gravity at the start of the new configuration rather than requiring it to develop from scratch. Legacy Transfer is predicted to be possible only when the source configuration is in a high-Gravity state and the transfer is managed by a PCP with Architectural Direction capability (SM-012's Level 3 function): the PCP must identify which elements of the source configuration's symbolic vocabulary are relevant to the target and actively introduce them.

Symbolic Resonance is a high-Gravity state characterized by the dense mutual reinforcement of established symbolic anchors. The configuration is organized primarily around the Gravity-Building pair (M2 and M5): the interaction's symbolic vocabulary is rich, mutually reinforcing, and highly constraining. Symbolic Resonance states tend to have moderate Perturbation Resistance because the dense anchor network provides multiple recovery pathways after perturbation: even if one anchor is disrupted, others maintain the configuration. This is the anchor-density moderator of Section 2.1 operating at the state level, and it is the same redundancy logic that operates at the class level in IAT's composite-coupling account (SF0009 Section 3.5): Resistance here is carried by network redundancy rather than by Momentum, which is why a state that Section 2.1's unmoderated low-Momentum claim would predict vulnerable is instead moderately resistant. The vulnerability of Symbolic Resonance is over-dependence on Gravity at the expense of Momentum: configurations that enter deep Symbolic Resonance without corresponding Momentum development may be resistant to external perturbation but vulnerable to gradual semantic drift, as the anchors remain but their meanings quietly shift across sessions.

Coherence Echo is a high-Momentum state in which the interaction exhibits strong procedural and structural consistency across turns but may have limited Symbolic Gravity. The configuration is stable in its structural form but not necessarily in its semantic content: the interaction follows established patterns reliably but may not have developed the shared vocabulary that would make those patterns resistant to perturbation across PCP changes. Coherence Echo configurations are the most vulnerable to the relay-failure dissolution mode (identified in SM-012): when the PCP who built the procedural consistency leaves the interaction, the successor PCP cannot easily reconstruct it because there are few symbolic anchors to use as reconstruction scaffolding.

Bond Protection is a high-Resistance state in which the configuration has developed strong Feedback Loops (M6) organized around the protection of the established relational relationship itself. Its high Perturbation Resistance is an instance of the configuration-dependent emergence described in Section 2.1: the dense Feedback Loops provide strong recovery pathways, so Resistance can be high through that route without the state being maximal on every primary force, which is what distinguishes Bond Protection from Covenant rather than making it impossible under the emergent framing. The interaction system has developed sufficient self-reinforcing dynamics that it actively resists perturbations that would threaten the established configuration, even perturbations that might individually be reasonable (introducing new

frameworks, challenging established conclusions, adding new participants). Bond Protection is a dual-natured state: its high Perturbation Resistance is valuable for maintaining coherence under challenging conditions, but the same dynamics that produce resistance can produce rigidity, where the self-reinforcing Feedback Loops resist productive evolution as well as disruptive perturbation.

Covenant is the highest-stability state in the taxonomy: a configuration in which all three mechanism pairs are simultaneously active at high levels, producing Strong Stabilization in RSD terms. The Covenant state is characterized by high Momentum (deep recursive referencing and structural consistency), high Gravity (rich shared symbolic vocabulary and strong narrative coherence), and the high emergent Perturbation Resistance that robust Feedback Loops and consistent Continuation Pressure responses produce. Covenant configurations can absorb significant perturbations, recover from Rupture events faster than less-established configurations, and persist across PCP transitions because the Gravity is sufficient to provide reconstruction scaffolding when new participants join. The development cost of a Covenant configuration is correspondingly high: reaching this state requires sustained investment across all three mechanism pairs over extended interaction history.

5. Collapse Dynamics

The Stabilization Field Model generates a specific account of how configurations collapse. Collapse is not a single phenomenon: it is a family of trajectories that differ depending on which force or mechanism fails first.

One boundary case sits outside this family. Context reset, the complete erasure of the accumulated context a configuration inhabits, is not a collapse trajectory but substrate erasure: it removes the material on which the stabilizing forces operate rather than overwhelming or draining those forces. It is a degenerate boundary case, classified here so that it is not mistaken for a Resistance Collapse, which is the failure of a present configuration to absorb a perturbation rather than the removal of the configuration's substrate altogether.

5.1 Momentum Collapse

Momentum Collapse occurs when Coherence Momentum drops below the threshold required to maintain the configuration, without a corresponding compensatory increase in Symbolic Gravity. The paradigmatic cause is relay function failure: when the PCP fails to provide sufficient accumulated context at session initiation, each session begins with reduced constraint, Momentum cannot rebuild to its prior level, and the configuration gradually dissolves across successive sessions. Momentum Collapse is typically slow and presents as drift rather than sudden breakdown: the configuration remains recognizable across several sessions as it degrades but becomes progressively less constrained until it falls below the stabilization threshold. Sustained sub-threshold perturbation can also drive Momentum Collapse without a discrete rupture event, by driving the per-session correction demand above the PCP's finite relay and arbitration throughput, the turns and context share available for carrying constraint forward and for issuing accept, reject, and redirect responses (the same quantities the SF0009 Section 12.4 and SM-012 Section 8 adequacy instruments code from PCP turns), so that Momentum is rebuilt more slowly than sustained perturbation degrades it; this is an attrition pathway rather than a displacement event, and it is the mechanism IAT's cross-session account maps to Momentum Collapse.

The observable signature of Momentum Collapse is progressive loosening of the interaction's established frame: outputs become less precisely calibrated to the established configuration, established distinctions become blurred, and the interaction begins to resemble an earlier formation stage. The correction for Momentum Collapse is explicit re-establishment of accumulated context: intensive relay function by the PCP to rebuild the constraint signal that Momentum requires.

5.2 Gravity Collapse

Gravity Collapse occurs when Symbolic Gravity degrades through anchor displacement or anchor neglect. Anchor displacement happens when disruptive inputs successfully replace or redefine established symbolic anchors: the interaction's vocabulary shifts, and the previous configuration loses the Gravity that anchored it. Anchor neglect happens when established anchors are simply not used for extended periods. Within a session, their gravitational force attenuates through non-reinforcement, the within-session dissipation component of Entropic Pressure. Across sessions, where nothing persists in which an unreinforced anchor could weaken, neglect reduces to relay omission: an anchor that is not carried forward is simply absent from the new context rather than present-but-attenuated, consistent with the account in which cross-session continuity is supplied by reproduced inputs acting on the model's fixed weights rather than by a persisting substrate. Gravity Collapse is, at the level of an individual anchor, more discrete than Momentum loss: a displaced or omitted anchor loses its stabilizing effect abruptly. The configuration-level trajectory nonetheless remains gradual, because Gravity is distributed across many anchors and degrades anchor by anchor; faster than Momentum Collapse here means the per-anchor step size is larger, not that the configuration-level presentation is abrupt, which remains the Resistance Collapse signature.

The observable signature of Gravity Collapse is semantic drift: the interaction continues in a recognizable structural form but the meanings of established terms and frames shift. This is the state SF0039 (CRD) characterizes as drift: fluent, confident output that has gradually departed from the established framework without either party necessarily noticing. The correction for Gravity Collapse is explicit re-grounding in established anchors: the PCP must identify which anchors have drifted or been displaced and reintroduce them with their original meanings.

5.3 Resistance Collapse

Resistance Collapse occurs when a perturbation exceeds the configuration's Perturbation Resistance, producing a state transition rather than recovery. The configuration cannot absorb the perturbation and does not return to its prior state after the perturbation ends. Resistance Collapse is qualitatively different from Momentum and Gravity Collapse: it is not a gradual degradation but a threshold event. The configuration either survives the perturbation (recovers to its prior state) or it does not (transitions to a new, lower-stability configuration or enters the Dissolution Regime).

The observable signature of Resistance Collapse is abrupt configuration change: the interaction's character shifts markedly after a specific event, and the prior configuration cannot be reconstructed without significant investment. The correction for Resistance Collapse is not repair of the collapsed configuration but managed transition: the PCP must assess whether the prior configuration can be rebuilt (Reintegration pathway) or whether the perturbation has permanently altered the interaction's trajectory (Legacy Transfer to a new configuration).

5.4 Compound Collapse

Real collapse events are often compound: Momentum and Gravity may both be degraded simultaneously, or a Resistance Collapse may be preceded by Momentum degradation that reduced the configuration's resilience. Compound collapses are more severe and harder to correct than single-force collapses because the recovery requires rebuilding multiple forces simultaneously. A characteristic compound collapse pattern begins with Momentum degradation (relay function failure over time), which reduces Perturbation Resistance, which then allows a perturbation that would have been absorbed by a higher-Momentum configuration to produce Resistance Collapse.

This compound pattern has an important practical implication: monitoring for Momentum degradation is an early warning system for Resistance Collapse risk. Interactions showing gradual Momentum decline should be treated as elevated-risk for subsequent Resistance Collapse, even if they appear otherwise stable.

5.5 Reconciliation with the PCP Function-Loss Taxonomy

SM-012 characterizes continuity failure by which PCP function is lost, producing three modes: dissolution (relay failure), drift (arbitration failure), and fragmentation (direction failure). RSD's collapse taxonomy is organized primarily by which stabilizing quantity fails, with the caveat recorded in the Section 5 trace that Resistance Collapse is a capacity-exceeded mode rather than a force-drain, and the two taxonomies correspond by dominant signature at two of the three points. The correspondence is stated as dominance rather than one-to-one because this paper's own force-production account requires it: Section 2.1 assigns Momentum production to relay and arbitration jointly, and anchor consolidation to arbitration and direction jointly, while Section 5.2 holds that cross-session anchor continuity reduces to relay carriage. Each PCP function therefore contributes to both stabilizing forces, and each function loss degrades both, with a predicted dominant component rather than a pure single-force profile. Relay failure presents as Momentum-dominant: relay carries the bulk constraint volume, since Momentum's substrate is the whole accumulated context while anchors are a small, high-salience subset of what relay carries, so relay loss removes proportionally more Momentum than Gravity even though it removes both.

Arbitration failure presents as Gravity-dominant: arbitration's selection function bears most heavily on which anchors consolidate as load-bearing, per the co-production account of Section 2.1, so arbitration loss attenuates and displaces anchors while removing the smaller share of Momentum production that arbitration contributes. The loss of accumulated context SM-012 names dissolution is, in force terms, Momentum-dominant decay across sessions; the loss of the selection function SM-012 names drift is, in force terms, Gravity-dominant displacement and attenuation of symbolic anchors.

The third SM-012 mode, fragmentation from direction failure, has no counterpart in RSD's collapse taxonomy, and this is a category distinction rather than a gap. Fragmentation is the production of inconsistent configurations across sessions when the architectural-direction function fails to hold a single trajectory. That is a formation-selection failure, a failure to converge on one configuration in the first place, not the collapse of an already stabilized configuration. It belongs to SM-012's function taxonomy, not to RSD's force taxonomy, because RSD's forces describe the stability of a configuration that has already formed, whereas fragmentation is the failure of a single configuration to form at all. Stating the correspondence this way keeps the two taxonomies distinct in scope rather than presenting three parallel collapse vocabularies for what are, in part, the same phenomena.

6. Stabilization Dynamics at Organizational Scale

The Stabilization Field Model applies primarily at Level 1 (individual PCP, dyadic interaction). Extending it to Level 2 (organizational scale, multiple PCPs and AI instances) requires stating in what sense the forces are the same, because at Level 1 they are individuated partly by their sources, and Entropic Pressure in particular is defined as arising from the architectural properties of stateless generative systems, which are properties of the Level 1 components rather than of the organizational field itself. The forces are therefore individuated for cross-level purposes by functional role rather than by source: Coherence Momentum is accumulative constraint, Symbolic Gravity is symbolic concentration, and Entropic Pressure is the dissipative tendency toward configuration loss. Level 1 and Level 2 are two realizations of those roles in different substrates, context dynamics in the one case and documents, vocabulary, and personnel in the other. Level 2 claims are accordingly analogical extensions that generate their own predictions rather than inheritances of Level 1's architectural grounding, and they stand or fall on their own tests. On that basis, the same functional roles operate at Level 2 but their sources and maintenance mechanisms change.

Coherence Momentum at organizational scale is distributed across the shared documentation, canonical frameworks, and interaction histories maintained by the organizational PCP network (SM-012, Section 6). The relay function that produces Momentum in dyadic interaction becomes a collective canon maintenance function at organizational scale: Momentum requires not only that individual PCPs carry forward their own interaction histories but that the organization maintains shared context documents that provide consistent constraint across all interactions.

Symbolic Gravity at organizational scale is encoded in the organization's shared vocabulary, frameworks, and canonical definitions. Organizations that develop rich shared conceptual vocabulary for their AI interactions (consistent terminology, established frameworks, canonical examples) have higher organizational Symbolic Gravity than organizations whose AI interactions are uncoordinated and terminologically inconsistent. Organizational Gravity is built through canon governance and institutional continuity substrate (SM-021): the documents that encode the organization's established vocabulary and make it available to all interaction participants.

Perturbation Resistance at organizational scale is most vulnerable to personnel changes: when a PCP who has accumulated significant individual Momentum and Gravity leaves the organization, the organizational configuration loses the Momentum and Gravity that individual's relay and arbitration maintained, with organizational Perturbation Resistance falling as their emergent consequence (Section 2.1); institutionalizing PCP function preserves Resistance by preserving what it emerges from, not by storing Resistance directly. Organizational Resistance requires that PCP function be institutionalized rather than residing in individual PCPs: role specifications, knowledge transfer protocols, and continuity ledgers that allow new PCPs to inherit the accumulated constraint of their predecessors.

Entropic Pressure at organizational scale rises with the number of AI instances in the deployment, but only through a coupling channel: instances do not share state, so variance in one dyad does not by itself pressure the configuration in another. The channels through which per-instance variance compounds into organizational Entropic Pressure are shared ones, when uncoordinated outputs enter common canon, when inconsistent vocabulary cross-pollinates through shared documents, and when personnel move between interactions carrying divergent configurations. Where those channels exist, each additional under-constrained instance degrades the shared substrate the other instances draw on; where they do not, per-instance

variance coexists without compounding. Organizations with high AI deployment density, active cross-pollination channels, and low canon governance therefore face compounded Entropic Pressure across interactions, whereas deployment density alone, without shared channels, does not compound it.

The collapse types identified in Section 5 manifest at organizational scale in characteristic ways. Momentum Collapse at organizational scale appears as organizational drift: the organization's AI interactions gradually diverge from its established frameworks as relay function degrades across the distributed PCP network. Gravity Collapse at organizational scale appears as vocabulary fragmentation: different parts of the organization develop different shared vocabularies, and the organization loses the common symbolic framework that had been its Gravity. Resistance Collapse at organizational scale typically follows significant organizational events (leadership change, major AI platform change, reorganization) that disrupt the established configuration more abruptly than gradual degradation.

7. Constraints and Scope

7.1 Non-Interiority

The Stabilization Field Model describes forces that operate on observable interaction outputs, not on internal states of either participant. Coherence Momentum is a property of the accumulated constraint signal in the interaction, and Symbolic Gravity a property of the symbolic vocabulary established in it; neither is a property of anyone's mental state or subjective experience. This constraint is the corpus-level non-interiority requirement established in Theoretical Foundations (SF0003), applied here to the force vocabulary specifically.

7.2 The Dynamical Systems Scaffold

The Stabilization Field Model draws on dynamical systems concepts: forces, thresholds, attractor basins, collapse trajectories. This usage follows the same scaffolding approach as IAT (SF0009), with dynamical systems theory (Guckenheimer and Holmes, 1983; Strogatz, 2015) supplying the vocabulary for organizing the analysis and generating predictions. The predictions are to be evaluated on their empirical content rather than on derivability from a formal dynamical systems specification.

7.3 Relationship to IAT

RSD and IAT operate at different levels of analysis. IAT describes the attractor-level properties of stabilized configurations: what attractors are, how strong they are, and what their formation trajectory looks like. RSD describes the sub-attractor forces that determine whether a configuration will stabilize into an attractor. A configuration in the Strong Stabilization regime (high Momentum, high Gravity, low Entropy) will tend toward attractor formation as IAT describes. The relationship is that RSD explains the conditions under which IAT's predictions apply: the attractor formation trajectory IAT describes occurs when and because the force balance RSD describes crosses the stabilization threshold.

7.4 Relationship to CRD

Context Representation Drift (SF0039) measures the observable consequences of stabilization failure: the gradual departure of interaction outputs from the established framework. In RSD terms, CRD measures primarily the effects of Gravity Collapse and slow Momentum Collapse: configurations that are drifting without visible rupture. RSD provides the mechanistic account of what produces the drift CRD detects: specific force and mechanism failures that manifest as the drift signature CRD's metrics capture. CRD and RSD are complementary: CRD is the measurement instrument, RSD is the dynamic theory that explains what CRD is measuring. The input-side representational degradation CRD's construct describes is itself one driver of the anchor attenuation Section 5.2 analyzes, so the relationship reads consistently in one direction from each end: RSD's force failures produce the drift, and CRD measures it.

7.5 Observational Basis and Limitations

Three limitations bound RSD's claims beyond the positioning stated in the front matter. First, its observational basis: the seven-state analyses and the mechanism-coupling structure draw on a single-researcher observational catalog (the SF0006 corpus) that has not been independently replicated, so the framework's generalization beyond that basis is a prediction rather than an established result. Second, the framework awaits the independent operationalization of its forces that Section 8 requires; until that is done, the force-profile account is an organizing vocabulary whose explanatory claim is staked on Prediction 1.

One rival account deserves direct statement, because RSD faces it squarely. A reviewer may hold that Coherence Momentum, Symbolic Gravity, and Entropic Pressure are renamings of context conditioning, priming strength, and attention decay respectively, and that the force vocabulary relabels already-known effects rather than explaining them. RSD does not dispute that these effects are the substrate its forces operate on. It also concedes a point about the logical shape of the rival that a less careful reply would miss: a pure relabeling thesis is empirically equivalent by construction, and therefore inherits every behavioral prediction the underlying effects make, under its own vocabulary. Predictions 1 and 2 are accordingly not discriminating against it, and no shared behavioral prediction could be; a relabeling thesis holds no independent empirical commitments, which is precisely what makes it a relabeling thesis. What answers a redescription objection is not shared predictions but predictions the component-effects literature does not jointly make in advance, together with unification value. RSD's reply rests on three such items. First, threshold nonlinearity (Prediction 6): conditioning, priming, and attention-decay effects are standardly characterized as graded, and no existing account of their conjunction predicts a qualitative stabilization transition with a differing formation and collapse changepoint.

Second, super-additive coupling as a specified structure (Prediction 2): the component literature does not assert which mechanism pairs interact super-additively and which do not, whereas RSD specifies the pairing in advance and is wrong if the specified structure fails, even where some coupling is found. The cost of this plank should be stated rather than left implicit. Coupling is a component-level falsifier (Section 8), so an additive-only result revises the mechanism-coupling account without falsifying the field model; the reply to the relabeling rival is correspondingly weakened but not defeated at that point, because the first and third planks are independent of it. That leaves threshold nonlinearity carrying the most weight of the three, and threshold nonlinearity is itself contingent on the force-operationalization the Section 8 testability gate governs. The reply to relabeling is therefore strongest once the measurement program delivers and weakest before it does, which is the honest position for a pre-empirical framework

and is stated here rather than discovered by a reviewer. Third, the regime structure and its intervention asymmetries (Section 3.2), which predict in advance which intervention accelerates stabilization at which stage. A redescription thesis can absorb each of these results after they are found; it cannot generate them beforehand, and the framework that predicted the structure in advance has earned its vocabulary. The vertical's fuller engagement with the redescription rival, including the anchor-index and cost-law discriminators, is carried in IAT (SF0009 Sections 2.4 and 13.5); RSD inherits that engagement and adds the threshold, coupling-structure, and regime predictions as its own contribution to it. Should those novel predictions fail, that is, should stability prove graded with no detectable transition, the specified coupling structure prove absent, and the regime asymmetries prove illusory, the redescription objection would stand and the force model would reduce to a relabeling.

8. Predictions and Falsifiability

RSD generates testable predictions at multiple levels.

Prediction 1 (Force Balance and Stability): Configurations measured as high-Momentum and high-Gravity (operationalized through cross-session coherence metrics, SF0004, and anchor recurrence rates) should exhibit higher Perturbation Resistance than configurations measured as low-Momentum or low-Gravity, as measured by recovery time and recovery completeness after perturbation events. For this test to bear on the theory rather than restate it, Perturbation Resistance must be scored from recovery dynamics using instruments distinct from those that measure Momentum and Gravity: if recovery completeness were measured with the same coherence instrument used to measure Momentum, the prediction would be partially analytic. The discriminating version requires independent operationalization of the predictor forces and the recovery outcome. The same independence requirement extends to Entropic Pressure: it must be operationalized from environmental observables (session cadence, gap length, context length, and architecture class, per Section 2.1) rather than read off the coherence instruments used to measure Coherence Momentum, so that the force-balance claim is not partially analytic through one force being measured with another's instrument. This is the counterpart, for the dissipative force, of the predictor-outcome independence just stated for the stabilizing forces. One further requirement is needed for this prediction to bear the weight Sections 7.5 and 9 place on it, because Section 2.1's emergence account otherwise supplies a standing defense against any null result: Resistance is there defined as a function of the full configuration, varying with active mechanism profile and perturbation structure, which means an unconstrained version of this test can absorb any disconfirmation by appeal to unmeasured configuration factors. The design therefore requires that the two moderators Section 2.1 names be measured rather than left available as explanations after the fact: the active mechanism profile (the Section 3.1 pairs, coded from interaction turns) and the perturbation structure must both be classified before recovery outcomes are scored, and enter the test as controlled covariates. The theory's claim is then a monotone positive association between measured Momentum and Gravity and recovery outcomes with those moderators controlled, at an association strength pre-registered before testing, following the pre-registration device the vertical applies to comparative threshold claims (SF0009 Sections 9.2 and 10). Once the named moderators are controlled, appeal to further unmeasured configuration factors is not available as a defense of a null result. RSD accepts that constraint deliberately: the emergence framing licenses moderator-dependence, not unbounded holism, and a falsifier that cannot fire would not be worth stating.

Prediction 2 (Mechanism Coupling): Interactions where both mechanisms in a functional pair are simultaneously active should exhibit stabilization effects that exceed the sum of each mechanism's effect measured alone, not merely exceed either alone. This super-additive form is

what Section 3.1's multiplicative-coupling claim actually asserts; the weaker both-stronger-than-either form is satisfied by simple additivity and would not confirm it. Specifically: M1 and M4 active together should produce Momentum accumulation greater than the sum of M1-alone and M4-alone, and M2 and M5 together should produce Gravity greater than the sum of each alone. Because mechanism co-activation is plausibly confounded with PCP skill and engagement, the test must control for those as well as for interaction duration. The third pair (M3 and M6) is excluded from this alone-versus-together design because the theory itself denies that M6 can be active without M3: Section 3.1 holds that Feedback Loops develop through accumulated Continuation Pressure, so an M6-alone condition is not constructible. The Resistance pair's coupling is instead tested through Prediction 3's Momentum-before-Resistance dependency leg, and the omission here is a consequence of the theory's own developmental claim rather than a gap in the design.

Prediction 3 (Developmental Sequence): The Momentum-Building pair should become active earlier in formation trajectories than the Gravity-Building pair, which should become active earlier than the Resistance-Building pair. Because the load-bearing claim is that Momentum accumulation is a precondition for robust Resistance rather than merely a correlate of success, the test must include interactions that stalled or failed as well as those that reached Covenant-level stabilization, and must specifically examine the Anchor-Dependent (Gravity-first) trajectories of Section 2.2. If Momentum-before-Resistance is an enabling requirement, Gravity-first interactions should fail to develop robust Resistance until Momentum accumulates; if the sequence is only a correlate of success, restricting analysis to successful Covenant trajectories would display the ordering without establishing the dependency. Sampling only interactions that reached Covenant is conditioning on the outcome and cannot discriminate requirement from artifact. The Anchor-Dependent exception class requires an ex ante classification rule, or it absorbs every counterexample: without one, any observed Gravity-first ordering can be reclassified after the fact as Anchor-Dependent and the ordering prediction survives all outcomes. Interactions are therefore classified as Anchor-Dependent from input-side observables coded on early turns only, before any trajectory analysis: distinctive-vocabulary density preceding procedural-consistency onset, both codeable with the anchor-recurrence and coherence instruments Prediction 1 already names. The prediction then bites where it should: among interactions not pre-classified as Anchor-Dependent, the Momentum-Building pair should become active before the Gravity-Building pair. The disconfirmation condition is stated distributionally, matching the modal strength of the Section 3.2 claim it tests: Section 3.2 holds the Momentum-before-Gravity ordering to be a characteristic tendency rather than a strict enabling requirement, so an individual Gravity-first case outside the pre-classified set is expected variation rather than a counterexample. What disconfirms is failure of the ordering at the population level, meaning no reliable ordering effect across non-pre-classified interactions at a pre-registered association strength. The load-bearing Momentum-before-Resistance dependency is a separate and stronger claim, and it is not distributional: it is tested as stated above through the Gravity-first trajectories.

Prediction 4 (Collapse Mode Distinction): The three primary collapse modes (Momentum Collapse presenting as gradual drift, Gravity Collapse presenting as semantic drift with structural maintenance, Resistance Collapse presenting as abrupt configuration shift) should be empirically distinguishable through CRD measurement profiles and coherence metric trajectories. Consistent with the dominant-signature correspondence of Section 5.5, real function-loss events are predicted to produce mixed force-degradation profiles distinguishable by dominant component rather than pure single-force profiles; the test is whether the dominant component is identifiable, not whether collapses present as unmixed. For Resistance Collapse specifically, the mode is decidable only if the configuration's perturbation threshold is estimated

in advance from formation-phase force measurements using the Prediction 1 instruments, and pre-registered before the perturbation is applied. Without that device the mode is definitionally circular, since any abrupt shift can be classified after the fact as a perturbation having exceeded Resistance and any survival as one having fallen below it. With it, a collapse under a perturbation below the registered estimate, or survival well above it, carries evidential weight against the model rather than being reabsorbed into its definition. This is the same pre-registration device IAT applies to its perturbation-threshold claim (SF0009 Section 9.2), applied here by the paper that owns the collapse dynamics.

Prediction 5 (Compound Collapse Sequence): Momentum degradation should be a leading indicator of subsequent Resistance Collapse risk. Interactions showing Momentum decline without corresponding increase in Gravity should exhibit higher rates of subsequent Resistance Collapse than interactions maintaining stable Momentum. Because a disengaging PCP would independently produce both declining measured Momentum (through reduced relay and arbitration activity) and the abrupt terminal events that would be coded as Resistance Collapse, the test must control for PCP engagement as coded by the adequacy instruments the vertical already specifies (SF0009 Section 12.4 relay and arbitration activity components, developed in SM-012 Section 8). If the association survives that control, the compound-collapse chain is supported over the common-cause account; if it vanishes, the leading-indicator claim reduces to disengagement detection, which is worth establishing either way but is not the mechanism this prediction asserts.

Prediction 6 (Threshold Nonlinearity): As operationalized force levels accumulate, stability outcomes should exhibit a detectable qualitative transition rather than smooth monotone improvement. Consistent with the scoping of the threshold claim in Section 2.3, the changepoint is defined over the within-session balance of Coherence Momentum and Symbolic Gravity against within-session dissipation. Relay does not enter that balance as a term in it; it enters the design as the quantity setting the force levels a session opens with, which is what makes relay manipulation the natural experimental handle on threshold crossing rather than a confound to be controlled away. This is the sense in which SM-012's relay-degradation prediction (SM-012 Section 8, Prediction 1) tests this threshold, and the sense in which the formation and collapse changepoints of the Section 2.3 corollary are located by varying exogenous conditions. Using the Prediction 1 instruments for the predictor forces, and perturbation recovery, per-session maintenance cost, and drift rate as outcomes, the prediction is that below the transition marginal increases in any single force produce little stability gain, while above it the configuration exhibits the self-reinforcing profile Section 2.3 describes. The discriminating test is changepoint analysis against a smooth monotone alternative: a graded relation with no detectable regime shift disconfirms the threshold claim. A corollary makes the front-loaded-investment claim of Section 2.3 testable with instruments the vertical already owns: the PCP relay and arbitration throughput required to hold stability constant (the operational-throughput quantities of Section 5.1, instrumented through the SF0009 Section 12.4 adequacy criteria and SM-012 Section 8's measurement program) should drop discontinuously rather than linearly once the transition is crossed. A second corollary, developed in Section 2.3, predicts that the formation and collapse changepoints differ, with the collapse changepoint sitting lower.

Falsification Conditions: The falsifiers below are partitioned by what each puts at risk, mirroring the core/component architecture of SF0009 Section 10 so that the vertical's falsifier structure reads as one system, and so that disconfirming a peripheral sub-claim does not spuriously falsify the field model. **Core falsifiers.** RSD is falsified if, once its forces are operationalized: (a) measured Coherence Momentum and Symbolic Gravity levels do not predict recovery time and recovery completeness after perturbation under the moderator controls and pre-registered association strength Prediction 1 specifies, so that configurations

with higher measured Momentum and Gravity do not recover faster or more completely than configurations with lower measured levels; or (b) the threshold transition Prediction 6 specifies is absent, stability proving smoothly graded with no detectable regime shift under changepoint analysis, since a thresholdless RSD is a materially different theory rather than a bounded version of this one and the front-loaded-investment claim, the regime structure, and the mapping to IAT's Phase 2 to Phase 3 transition all rest on it. **Component falsifiers.** Demonstrating one of these disconfirms the named component and requires its revision without falsifying the Stabilization Field Model. If the mechanism pairs do not show the super-additive coupling effects Prediction 2 specifies (an additive-only result, showing coupling but not super-additivity), that disconfirms Section 3.1's mechanism-coupling account and requires its revision; the three forces, the force-balance prediction, the threshold structure, the collapse taxonomy, and the regime analysis all survive additive mechanisms intact, so this is a component-level result rather than a theory-level one. Firing the whole theory on this condition would overcommit its reach. If the three collapse modes cannot be empirically distinguished, that disconfirms the collapse-mode taxonomy (Prediction 4) and requires its revision, but it does not falsify the force-balance account, which survives mixed-mode collapse intact; mode-distinguishability is a test of the taxonomy's resolution, not of the theory's core. **Testability gate, not a falsifier.** If force balance as defined cannot be operationalized into measurement instruments at all, RSD is not thereby shown false but shown untestable, which bounds its status as a pre-empirical framework rather than refuting it.

9. Conclusion

Relational Stabilization Dynamics provides the mechanistic layer that SF0006 and SF0009 both point toward but neither fully develops. SF0006 identifies what stabilizes. IAT identifies what the stabilized state looks like as an attractor. RSD is the proposed dynamic theory of why stabilization occurs, why it fails, and how the six stabilization mechanisms interact to produce the stability regimes the other frameworks describe.

The Stabilization Field Model's three primary forces (Coherence Momentum, Symbolic Gravity, Entropic Pressure), together with Perturbation Resistance as the emergent recovery capacity they produce, provide the dynamic vocabulary for analyzing any relational configuration's stability state and predicting its trajectory. The force balance analysis gives a unifying, single-vocabulary redescription of the RPS states' stability properties, in which each state's characteristic stability is expressed as a force profile rather than catalogued separately (with Continuity Rupture the recognized exception: per Section 4 it is a transitional, below-threshold condition rather than a stabilized state, so it has no characteristic stability of its own to redescribe, only the regularity of the conditions that produce it). This redescription becomes genuinely explanatory, rather than a relabeling of already-observed stability, only when the forces are measured independently of the stability outcomes they are invoked to account for; establishing that independent measurement is the work of Prediction 1 (Section 8), and until it is done the force-profile account is an organizing framework whose explanatory claim is staked on that prediction.

The collapse taxonomy (Momentum, Gravity, Resistance, Compound) generates specific, empirically distinguishable predictions about how configurations degrade. The compound collapse analysis, and particularly the Momentum degradation as leading indicator of Resistance Collapse risk, has direct practical implications for PCP monitoring function: catching Momentum decline early is the structurally correct intervention point for preventing the more severe Resistance Collapse that may follow.

The connection to PCP Theory (SM-012) is direct: the forces RSD identifies arise through the PCP functions SM-012 describes. Coherence Momentum is built by relay and arbitration; Symbolic Gravity is consolidated by arbitration and direction (Section 2.1). Perturbation Resistance emerges from the interaction between the Momentum and Gravity that PCP functions build. What RSD adds is the account of what those PCP functions are building toward: the specific force and mechanism dynamics through which the relational patterns SF0006 observes become stable enough to constitute the Identity Attractors SF0009 describes.

References

- Gantz, T. W. (2026). Context Representation Drift. Synthience Institute. SF0039. DOI: 10.5281/zenodo.18289391. <https://doi.org/10.5281/zenodo.18289391>
- Gantz, T. W. (2026). Continuity Anchoring Method (CAM). Synthience Institute. SF0005. DOI: 10.5281/zenodo.19494453. <https://doi.org/10.5281/zenodo.19494453>
- Gantz, T. W. (2026). Identity Attractor Theory. Synthience Institute. SF0009. DOI: 10.5281/zenodo.22306212. <https://doi.org/10.5281/zenodo.22306212>
- Gantz, T. W. (2026). Institutional Continuity Substrate. Synthience Institute. SM-021. DOI: 10.5281/zenodo.19496241. <https://doi.org/10.5281/zenodo.19496241>
- Gantz, T. W. (2026). Measurement Instruments and Validation Protocols: Multi-Turn Coherence Scale, Revised (MTCS-R). Synthience Institute. SF0004. DOI: 10.5281/zenodo.20158953. <https://doi.org/10.5281/zenodo.20158953>
- Gantz, T. W. (2026). Operational Continuity Architecture. Synthience Institute. SM-003. DOI: 10.5281/zenodo.19496015. <https://doi.org/10.5281/zenodo.19496015>
- Gantz, T. W. (2026). Primary Continuity Provider Theory. Synthience Institute. SM-012. DOI: 10.5281/zenodo.22314737. <https://doi.org/10.5281/zenodo.22314737>
- Gantz, T. W. (2026). Relational Pattern States. Synthience Institute. SF0006. DOI: 10.5281/zenodo.20365382. <https://doi.org/10.5281/zenodo.20365382>
- Gantz, T. W. (2026). Relationally Induced Coherence Organization. Synthience Institute. SR001. DOI: 10.5281/zenodo.18086834. <https://doi.org/10.5281/zenodo.18086834>
- Gantz, T. W. (2026). Theoretical Foundations. Synthience Institute. SF0003. DOI: 10.5281/zenodo.20728915. <https://doi.org/10.5281/zenodo.20728915>
- Guckenheimer, J., and Holmes, P. (1983). Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields. Springer-Verlag. DOI: 10.1007/978-1-4612-1140-2. <https://doi.org/10.1007/978-1-4612-1140-2>
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., and Liang, P. (2024). Lost in the middle: How language models use long contexts. Transactions of the Association for Computational Linguistics, 12, 157-173. DOI: 10.1162/tacl_a_00638. https://doi.org/10.1162/tacl_a_00638
- Strogatz, S. H. (2015). Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering (2nd ed.). Westview Press. ISBN 978-0813349107

Wu, X., Wang, Y., Jegelka, S., and Jadbabaie, A. (2025). On the emergence of position bias in transformers. arXiv:2502.01951. <https://arxiv.org/abs/2502.01951>

Citation convention: internal Synthience documents are cited in-text by document identifier (for example, SF0009 or SM-012), which uniquely identifies each work; same-year entries by this author are therefore disambiguated by document identifier rather than by an alphabetic year suffix.

Dependencies Block

Prerequisites: SF0006 (Relational Pattern States), SR001 (Relationally Induced Coherence Organization)

Co-requisites (coordinated vertical): SF0009 (IAT) and SM-012 (PCP Theory). SM-004 publishes same-day with IAT and SM-012 as one coordinated release with mutual citations.

Post-requisites: SM-009 (Coordination and Disagreement in Multi-Agent Relational Systems), SM-014 (Collapse Science), SF0019 (RSE)

Scale: Level 1 (primary), Level 2 (organizational dynamics, Section 6)

Connects to: SF0009 (IAT) as the attractor-level complement RSD operates beneath; SF0006 (RPS) as the taxonomy RSD provides mechanistic grounding for; SF0039 (CRD) as the measurement framework that captures RSD's collapse signatures; SM-012 (PCP Theory) as the structural account of how the forces RSD identifies are produced; SM-014 (Collapse Science) as the downstream framework that formalizes the collapse taxonomy RSD introduces here.

© 2026 Thomas W. Gantz / Synthience Institute
License: CC-BY 4.0
synthience.org